

Spatial Storytelling and Scientific Data Visualization

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ABSTRACT

With misinformation spreading as quickly as the coronavirus itself, we saw an urgent need to illustrate complicated scientific concepts in a clear way to help readers assess risk and empower them to stay safe. We're going to dive into reporting/design/production and visualization of three pieces that were published recently.

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1 INTRODUCTION

An airborne virus that spreads through particles and droplets was a perfect candidate for 3-D and spatial analysis, and the subject matter allowed us to present the data in its true form using immersive storytelling on multiple platforms. Through WebGL and augmented reality, these stories demonstrated how infectious aerosols can move in spaces of vital interest to public health: within the fibers of a face mask and in a classroom.

2 PUBLISHED ARTICLES

The following are the published articles we'll be presenting in our session:

2.1 Masks Work. Really. We'll Show You How

This project allowed readers to see clearly the protection masks can provide, turning an abstract notion of safety into a tangible visual explanation.

The reporting was based on interviews with leading aerosol experts, and the backbone of the presentation was a 3-D animation [Fleisher et al., 2020] at the particle scale, which let readers see the individual fibers of a mask and an N95 respirator. The animation moved through the physics of filtration at work at a microscopic level to show that even though the material of these masks looks thin, they are able to trap viral particles effectively.

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2.2 Why Opening Windows Is a Key to Reopening Schools

As schools considered reopening [Bartzokas et al., 2021] for in-person learning, we built on our experience simulating particles and airflow to illustrate the simple steps schools could take to reduce coronavirus exposure.

We gained access to a Harlem classroom to take precise measurements and construct a 3-D model of the space. We worked with an engineering firm to show how air would move around the room under different scenarios, how infectious aerosols might build up in the space, and, crucially, how to reduce them.

Reporting with leading experts in ventilation and New York City's school reopening plan helped place the data in context.

2.3 See How the Dixie Fire Created Its Own Weather

Days after California's Dixie fire [Popovich et al., 2021] ignited in July 2021, towering storm clouds burst from the flames, generating lightning and wild winds. By merging data from radar stations in California and Nevada, we were able to reconstruct Dixie's first thunderclouds in 3-D. Terrain, fire, lightning and other data sets were then designed, layered together and rendered out, revealing how the wildfire fed its own destructive spread.

These stories helped communicate a vital public health issue in an engaging and visually impactful format, providing some clarity during a time of uncertainty and anxiety. Our unique visual approach helped connect millions of readers with emerging research in an intuitive, three-dimensional mode of interaction. We tried several approaches to reach audiences with this crucial public health information and made several technological leaps in our visual storytelling. The team developed a bespoke system to turn our spatial reporting and modeling into a story experience across multiple platforms: a clear, immersive 3-D experience for the web, a robust 2-D print graphic, and — for our Instagram audience — an augmented reality effect.

It is difficult to overstate the degree of technical difficulty this project presented and how inversely smooth the overall presentation was for readers. This represents a significant technological advancement for Times storytelling and the development work for these projects will provide a foundation for future projects.

Simulation data of thousands of particles tends to be very large and difficult to implement in a real-time, interactive environment. To bring it into our publishing framework, we developed a point cloud processing pipeline that read, filtered and stored the data in a custom file format, optimized for size and performance. This

pipeline allowed us to preview and animate the data in a 3-D editor, using color and opacity stream tracer animations to advance through the simulation over time using scroll. An efficient GPU rendering technique ensured a smooth experience for our readers across all platforms without compromising quality.

To make the impact of ventilation on coronavirus spread more tangible, we also used augmented reality on Instagram to go inside the fibers of a mask and to see the simulated particle flow in the classroom. In the mask A.R. experience, users become the size of a particle themselves to understand the mask fibers' ability to trap them. In the classroom experience, users can walk around as their phones become a Geiger counter-like measuring device reading contaminant levels.

These projects were collectively viewed millions of times and were among the most popular and distinctive pieces of coronavirus coverage published by The Times.

Sources: Audrey Dang and Brent Williams, Department of Energy, Environmental & Chemical Engineering and the Center for Aerosol Science and Engineering, Washington University in St. Louis; Linsey Marr, Department of Civil and Environmental Engineering, Virginia Tech; Joseph G. Allen, T.H. Chan School of Public Health, Harvard University; Chris Zangmeister, Material Measurement Laboratory, National Institute of Standards and Technology; Edward Vicenzi, Smithsonian Institution's Museum Conservation Institute; Christian J. Kähler, Institute of Fluid Mechanics and Aerodynamics, Bundeswehr University Munich; Jesse V. Jokerst, Department of Nanoengineering, University of California, San Diego; Frank Drewnick, Particle Chemistry Department, Max Planck Institute for Chemistry; Richard Stutt, Epidemiology and Modelling Group, University of Cambridge.

About this story

This simulation was created using a three-dimensional model of a New York City public school classroom, with specific inputs for the size of the room and the number of occupants and their location in the room, among other factors. Nearly 800,000 points of data were sampled from a simulation of airflow in the room. Two values — contaminant concentration (as parts per million) and velocity (as meters per second) — were measured for each point every few seconds over 15 minutes. The animation shows a subset of this data. The data was visualized both as streams and as horizontal cross-sections. The color range is scaled linearly from 0ppm to 5ppm.

Sources: Scott E. Frank and Gregory A. James, JB&B; Joseph G. Allen, T.H. Chan School of Public Health, Harvard University; Mark Thaler, Gensler.

Parham Azimi of the T.H. Chan School of Public Health and John Wilson of Siemens EDA provided technical assistance. How we built the 3-D mo

The Dixie fire's smoke and clouds were created using reflectivity data from three Next-Generation Radar network stations collected on July 19, 2021, between 11:00 a.m. and 11:00 p.m. Pacific Daylight Time. The raw data was collected every 10 minutes in radial sweeps around the radar stations, each at a higher altitude. The Times combined and reformatted the data using Py-ART, a collection of algorithms and utilities used regularly in radar analysis. We then filtered it to reduce noise.

We applied color and texture to the 3-D volume to approximate a smoke- and cloud-like look. And we interpolated the sequence in time to create a smoother video animation. The radar volume rendering was based on previous work by Dr. Lareau of the University of Nevada, Reno. The lightning strikes were recorded by the Vaisala Lightning Detection System. We only showed strikes that reached the ground.

The terrain was created using topographic data from the U.S.G.S. 3-D Elevation Program and color imagery from the Copernicus Sentinel-2 satellites, retrieved at a resolution of 10 meters. To eliminate 2-D cloud cover on the landscape, we generated a composite of satellite images from June 1 to June 30, 2021.

The fire footprint was created by combining perimeter data from the California Department of Forestry and Fire Protection along with fire detections data from NASA's Fire Information for Resource Management System and the U.S. Forest Service Active Fire Mapping Program.

Additional expert sources:

- Michael Fromm, meteorologist at the U.S. Naval Research Laboratory.
- Daniel Swain, climate scientist at the University of California, Los Angeles.
- Rich Thompson, incident meteorologist at the National Weather Service, Los Angeles/Oxnard.

Additional credits:

- Produced by Michael Beswetherick.
- Photo editing by Matthew McCann.
- Additional work by Or Fleisher, Sam Manchester and Jeremy White.
- Top video from ALERTWildfire and PG&E.

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